

Counting similar triangles with coordinates in finite fields

Corbin Smith
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In this essay we will count every non-null triangle with coordinates in a finite field of p elements, up to similarity.

Rational Trigonometry, introduced by N. J. Wildberger in *Divine Proportions*, replaces distances and angles with the algebraic notions of quadrance and spread, providing similarity invariants that remain valid over arbitrary fields and quadratic forms. In this framework, a non-null triangle over $(\mathbb{Z}/p\mathbb{Z})^2$ with $p \neq 2, 3$ is determined up to similarity by its spread triple (s_A, s_B, s_C) , and these triples satisfy the symmetric cubic relation

$$(x + y + z)^2 - 2(x^2 + y^2 + z^2) - 4xyz = 0, \quad xyz \neq 0.$$

Thus similarity classes correspond to $\mathbb{Z}/p\mathbb{Z}$ -rational points on this cubic surface modulo the natural S_3 -action.

Using the triple spread formula developed by Wildberger together with the orbit-counting lemma, we show that the number of similarity classes of non-null triangles over $\mathbb{Z}/p\mathbb{Z}$ is

$$\frac{p^2 - 1}{6}.$$

Despite the simplicity of this statement, I believe it is an original result. Computational evidence further suggests that the set of realizable spread triples decomposes into two disjoint families, determined solely by whether the determinant of the underlying quadratic form is a quadratic residue or non-residue modulo p .

Some algebraic derivations are left as exercises to keep the exposition concise. Occasional links to geometric algebra and the wedge product are noted but are not required for following the main argument. Every result here can be proved with the dot product alone, but the inclusion of bivectors, and their quadrances, significantly simplifies and informs this discussion. All greek letters are scalars, bolded letters are vectors, and upper case letters represent sets, points, surfaces, matrices, or functions. Familiarity with the combinatorics of the projective plane as well groups and fields is assumed.

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1 Geometric background

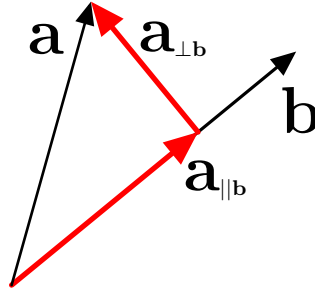
Suppose we have a vector space $\mathbb{V}$ over a field $\mathbb{F}$, equipped with a function $Q(\mathbf{x})$, called a *quadratic form*, such that $Q(\lambda\mathbf{x}) = \lambda^2 Q(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{V}$. We call the process of applying Q to $\mathbf{x}$ as *measuring* $\mathbf{x}$, and $Q(\mathbf{x})$ is the *quadrance* of $\mathbf{x}$. These conditions mean that when we have a basis then Q is a homogeneous quadratic expression in the coefficients of its argument, and therefore can be expressed as $Q(\mathbf{x}) = \mathbf{x}^T M \mathbf{x}$ for some unique symmetric matrix M . We refer to M as the symmetric matrix of our quadratic form. We call $\det(M)$ the determinant of our quadratic form. With $Q(\mathbf{x})$ and M we can define the *dot product* of $\mathbf{x}$ and $\mathbf{y}$ to be computed as $\mathbf{x} \cdot \mathbf{y} = (Q(\mathbf{x} + \mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}))/2 = \mathbf{x}^T M \mathbf{y}$, which is a symmetric bilinear expression in $\mathbf{x}$ and $\mathbf{y}$. We define $\mathbf{x}$ to be *perpendicular* to $\mathbf{y}$ exactly when $\mathbf{x} \cdot \mathbf{y} = 0$ because this makes the Pythagorean true with any dot product we pick. For any Q , we can determine the coefficients of M by making a finite number of measurements of all the basis vectors and their pairwise sums. For convenience we write $Q(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x}$ as $\mathbf{x}^2$. Any nonzero vector perpendicular to itself $\mathbf{v}^2 = 0$ is said to be a *null* vector. In this paper we will assume the characteristic of our field is not 2 and M is always invertible. The case of characteristic 3 is also treated separately.

Given two vectors $\mathbf{a}$ and non-null $\mathbf{b}$, we will want to be able to split $\mathbf{a}$ into a sum of two vectors $\mathbf{a}_{\parallel \mathbf{b}}$ and $\mathbf{a}_{\perp \mathbf{b}}$, called the projection and rejection of $\mathbf{a}$ onto $\mathbf{b}$, so that one is parallel to $\mathbf{b}$ and the other is perpendicular.

Exercise 1 Show that we can compute $\mathbf{a}_{\parallel \mathbf{b}}$ as

$$\mathbf{a}_{\parallel \mathbf{b}} = \frac{(\mathbf{a} \cdot \mathbf{b})}{\mathbf{b}^2} \mathbf{b},$$

and in geometric algebra notation, $\mathbf{a}_{\perp \mathbf{b}} = (\mathbf{a} \wedge \mathbf{b}) \mathbf{b}^{-1}$.



Given two (nonnull) vectors $\mathbf{a}$ and $\mathbf{b}$, we wish to come up with some notion of separation between them. One way that turns out to be symmetric is divide $\mathbf{a}_{\perp \mathbf{b}}^2$, the quadrance of the rejection of $\mathbf{a}$ onto $\mathbf{b}$, by $\mathbf{b}^2$. This is what is called the *spread* of the two vectors, denoted

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by $s(\mathbf{a}, \mathbf{b})$ and it can be computed as $s(\mathbf{a}, \mathbf{b}) = 1 - (\mathbf{a} \cdot \mathbf{b})^2 / (\mathbf{a}^2 \mathbf{b}^2)$ using the formula for a projection and the Pythagorean theorem.

Exercise 2 Show that in Euclidean geometry we have that $s(\mathbf{a}, \mathbf{b}) = \sin^2 \theta(\mathbf{a}, \mathbf{b})$. Confirm that $s(\mathbf{a}, \mathbf{b}) = s(\mathbf{b}, \mathbf{a}) = s(\delta \mathbf{a}, \epsilon \mathbf{b})$ whenever $\delta \epsilon \neq 0$. Show that in a plane where $\mathbf{a} = (x_a, y_a)$, $\mathbf{b} = (x_b, y_b)$, $(1, 0)^2 = \alpha$, $(1, 0) \cdot (0, 1) = \beta$ and $(0, 1)^2 = \gamma$ that

$$s(\mathbf{a}, \mathbf{b}) = \frac{(\alpha\gamma - \beta^2)(x_a y_b - x_b y_a)^2}{(\alpha x_a^2 + 2\beta x_a y_a + \gamma y_a^2)(\alpha x_b^2 + 2\beta x_b y_b + \gamma y_b^2)} = \frac{-(\mathbf{a} \wedge \mathbf{b})^2}{\mathbf{a}^2 \mathbf{b}^2}$$

for an explicit formula. Observe that the spread of two vectors is zero exactly when they are collinear.

Similar to spread we define the *cross* of two vectors as $c(\mathbf{a}, \mathbf{b}) = (\mathbf{a}_{\parallel \mathbf{b}}^2) / \mathbf{b}^2 = (\mathbf{a} \cdot \mathbf{b})^2 / (\mathbf{a}^2 \mathbf{b}^2)$. We introduce it here for completeness and to provide context for the first lemma.

For the remainder of the paper we will use the following conventions for a triangle ABC . Denote the side vectors by $\mathbf{a} = B - C$, $\mathbf{b} = C - A$, $\mathbf{c} = A - B$, and vertex spreads as $s(\mathbf{b}, \mathbf{c}) = s_A$, $s(\mathbf{c}, \mathbf{a}) = s_B$, and $s(\mathbf{a}, \mathbf{b}) = s_C$.

In a previous essay, we proved three formulas called the *spread law*, *cross law*, and *triple spread law* using geometric algebra, which Wildberger initially proved in his book *divine proportions* by geometric considerations. We will restate the first two, leaving their proofs as exercises to the reader, and then use them to prove the third

Lemma 1 *the cross law* Given a triangle ABC , then we have that

$$4\mathbf{a}^2 \mathbf{b}^2 (1 - s_C) = (\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2)^2.$$

Lemma 2 *the spread law* Given a triangle ABC we have that

$$\frac{s_A}{\mathbf{a}^2} = \frac{s_B}{\mathbf{b}^2} = \frac{s_C}{\mathbf{c}^2}.$$

To wrap your mind around these two lemmas, it is helpful to notice that they can be obtained as the squares of the law of cosines and law of sines from euclidean geometry. What is nice is that these statements are actually completely general, they hold for any quadratic form over any field not of characteristic 2. While the cross law can be obtained from squaring the right vector difference twice, the spread law can be easily proved with geometric algebra, using the theory of bivector quadrances.

Exercise 3 Show that the cross law can be rewritten as

$$(\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2)^2 = 2(\mathbf{a}^4 + \mathbf{b}^4 + \mathbf{c}^4) + 4\mathbf{a}^2 \mathbf{b}^2 s_C$$

Can the cross law be interpreted in terms of determinants of metric tensors and the quadrance of a bivector?

With these two results, we can prove the analog of the fact that the sum of the angles of a triangle add up to π .

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Theorem 1 *the triple spread law* Given a triangle ABC we have that

$$(s_A + s_B + s_C)^2 = 2(s_A^2 + s_B^2 + s_C^2) + 4s_A s_B s_C.$$

Proof. From the spread law we have that $s_A/\mathbf{a}^2 = s_B/\mathbf{b}^2 = s_C/\mathbf{c}^2 = 1/D$ for some nonzero constant D , which means we can say $Ds_A = \mathbf{a}^2$, $Ds_B = \mathbf{b}^2$, and $Ds_C = \mathbf{c}^2$. Substituting these expressions for the quadrances of our vectors into the alternate form of the cross law, and dividing by D^2 achieves the result. $\square$

For ease of exposition we denote the solutions of $(x + y + z)^2 - 2(x^2 + y^2 + z^2) - 4xyz = 0$ as the surface $\mathcal{E}$. Let T_p be the set of ordered solutions to $\mathcal{E}$ with $xyz \neq 0$ modulo p .

Exercise 4 Notice that the algebraic condition we have obtained is a cubic in the spreads. Show that we can rewrite the triple spread law as a quadratic equation in any of the three variables

$$s_C^2 - 2s_C(s_A + s_B + s_A s_B) + (s_A - s_B)^2 = 0$$

Conclude that if $s_A = s_B = s$ then either $s_C = 0$ or $s_C = 4s(1 - s)$. Additionally, if the characteristic of our field is not 2 or 3, show that the points $(0,0,0)$, $(1,1,0)$, $(1,0,1)$, and $(0,1,1)$ are the only singularities of $\mathcal{E}$, and $(3/4, 3/4, 3/4)$ is always a regular point of $\mathcal{E}$.

The last definitions we need are probably the most familiar to everyone, at least in spirit. A triangle ABC is said to be *non-null* when the points ABC are not collinear, and the vectors determined by the sides of the triangle are non-null as well. We say that two triangles ABC and DEF are *similar* whenever $s_A = s_D$, $s_B = s_E$, and $s_C = s_F$.

Notice that triangle similarity is an equivalence relation, and therefore partitions all triangles into equivalence classes. If we are working over a finite field, then there are a finite number of possible triangles and hence a finite number of classes. These classes correspond to elements (x, y, z) of the set T_p up to the natural permutation action of S_3 .

2 The orbit-counting lemma

To keep this essay concise we will not review the basics of group theory, but we will recall the orbit-counting lemma, which the reader is invited to prove. Given a finite set X , acted on by a group G , denote the set of orbits of the action by X/G . For each element $g \in G$, denote the set of fixed points of X by g as X^g .

Lemma 3 For a set X acted on by a finite group G , we have that

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

the number of orbits is the average of the size of the set of the points fixed by group elements.

3 The main result

We now turn to the task of counting all non-null triangles in $(\mathbb{Z}/p\mathbb{Z})^2$ up to similarity. Suppose $\mathbb{F} = \mathbb{Z}/p\mathbb{Z}$ our field is the integers modulo p , where p is a prime not equal to 2.

Exercise 5 Show that if $\mathbb{F} = \mathbb{Z}/3\mathbb{Z}$ then $|T_3/S_3| = 1$ by demonstrating that a non-null triangle can only be right and isosceles in this situation. Explain why there are no non-null triangles over this field when measured with the quadratic form $\mathcal{R}(x, y) = x^2 - y^2$, but we can find non-null triangles with the quadratic form $\mathcal{B}(x, y) = x^2 + y^2$.

The previous exercise opens the door to the fact that not every possible triangle similarity class can be realized by every quadratic form. To this end, we will be counting the number of possible similar triangles across all possible quadratic forms. A conjecture and some empirical observations about the distribution of equivalence classes across quadratic forms is given at the end.

Theorem 2 *Similar triangles mod p* If p is a prime number not equal to 2 or 3, then

$$|T_p/S_3| = \frac{p^2 - 1}{6}.$$

Proof The argument will proceed in two parts, first we will show that there are $p^2 - 3p + 3$ ordered triples corresponding to non-null triangles, and then we will count how many of them are fixed by each of the elements of S_3 the symmetric group of order 3.

For the first part of the argument, observe that because the origin is a singularity of $\mathcal{E}$ any line through the origin intersects $\mathcal{E}$ there with multiplicity at least 2. To analyze the intersections, we substitute $(\lambda a, \lambda b, \lambda c)$ into $\mathcal{E}$ which gives us a cubic equation in λ , with $\lambda = 0$ (as a double root) or

$$\lambda = \frac{2(ab + bc + ca) - (a + b + c)^2}{4abc}.$$

This means that for most projective points $(a : b : c) \in \mathbb{P}^2(\mathbb{Z}/p\mathbb{Z})$, we can find at least one solution in that direction on our surface. The exceptions correspond to lines with a triple intersection at the origin. There are three lines connecting the origin to the other singularities, but all such will stay in the coordinate planes, meaning we would automatically exclude those points since we are insisting that $abc \neq 0$. This restriction also means there are $(p - 1)^2$ directions in $\mathbb{P}^2(\mathbb{F}/p\mathbb{Z})$ we need to consider initially. Of these, we also would need $2(a^2 + b^2 + c^2) - (a + b + c)^2 \neq 0$. Since the conic $2(ab + bc + ca) - (a + b + c)^2 = 0$ has no singular points, it is a standard fact that it has $p + 1$ projective points. This conic is also tangent to the three coordinate planes, so of these $p + 1$ points, we have already excluded 3 of them, leaving us with $p - 2$ additional points we need to exclude. Therefore the number of directions in $\mathbb{P}^2(\mathbb{Z}/p\mathbb{Z})$ corresponding to elements of T_p is $(p - 1)^2 - (p - 2) = p^2 - 3p + 3$.

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Now we will use the orbit-counting lemma which tells us that

$$|T_p/S_3| = \frac{1}{|S_3|} \sum_{\sigma \in S_3} |T_p^\sigma|.$$

The symmetric group S_3 consists of an identity element, three transpositions, and two 3-cycles. The identity fixes all $p^2 - 3p + 3$ ordered solutions, and both of the 3-cycles fix points of the form (x, x, x) . Substituting this into $(x + y + z)^2 - 2(x^2 + y^2 + z^2) - 4xyz = 0$ gives $3x^2 = 4x^3$, so the only point satisfying this condition is $(3/4, 3/4, 3/4)$. The three transpositions each fix solutions with two entries equal: (x, x, y) , (x, y, x) , or (y, x, x) . Since this situation implies $y = 4x(1 - x)$, which is zero for only $x = 0, 1$, there are $p - 2$ such solutions. Putting all this together we have that

$$|T_p/S_3| = \frac{1}{6}(p^2 - 3p + 3 + 3(p - 2) + 2) = \frac{p^2 - 1}{6}$$

as required. □

This beautiful result is only the first of potentially many in this intersection between algebra, combinatorics, and geometry. A reasonable question is if any similarity class can be achieved by some quadratic form. Presently the author cannot answer this question. Based on looking at the empirical counts of the similarity classes over small finite fields I have arrived at a conjecture.

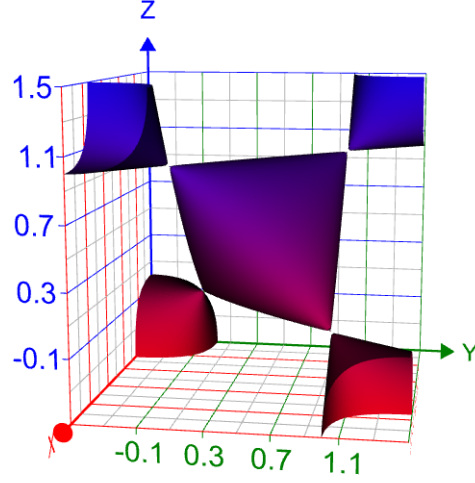
Conjecture *Determinantal partition* For each prime $p \neq 2$, we have that the set T_p/S_3 decomposes into the disjoint union

$$T_p/S_3 = T_p^+ \sqcup T_p^-$$

where all the unordered triples of T_p^+ are realized by every quadratic form whose determinant is a quadratic residue modulo p , and all the unordered triples of T_p^- are realized by every quadratic form whose determinant is a quadratic non-residue modulo p .

I can confirm that this conjecture is true for all primes $3 < p < 100$, by looking at all realized similarity classes of every possible quadratic form modulo p , and that $|T_p^+| \neq |T_p^-|$ in general. Empirically, two quadratic forms of the same determinantal type need not realize the classes with equal frequency.

Below is a render of a portion of $\mathcal{E}$, which Wildberger refers to as the *Ellipson*. The attentive reader may have noticed that the Ellipson's tangent cone at the origin, is exactly the numerator of λ in the proof of the main result.



If the conjecture $T_p = T_p^+ \sqcup T_p^-$ holds, the situation parallels the real case. In this setting, the Cauchy-Schwartz inequality tells us the points on $\mathcal{E}$ that can be realized by any quadratic form whose determinant is a positive number must be in the interior of the unit cube. By proving the reverse Cauchy-Schwartz inequality for quadratic forms whose determinant is a negative number, we can show that such quadratic forms can only realize points on $\mathcal{E}$ in the exterior of the unit cube.

Exercise 6 Show that if $\theta + \Phi + \Psi = \pi$ and $0 < \theta, \Phi, \Psi < \pi$ then $(\pi - \theta) + \Phi + \Psi = \pi$ is only possible if $\theta = \pi/2$, and both $(\pi - \theta) + (\pi - \Phi) + \Psi = \pi$ and $(\pi - \theta) + (\pi - \Phi) + (\pi - \Psi) = \pi$ are impossible. Conclude that spread similarity is equivalent to angle similarity, in euclidean geometry.

4 Examples

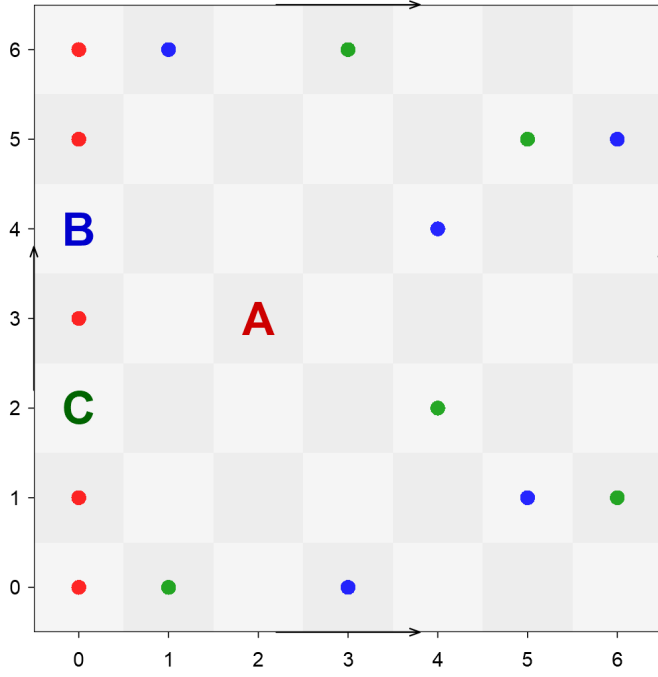
For the benefit of the reader, we will show some pictures of triangles over finite fields, and some tables of random forms over small fields, which detail the similarity classes they can realize. Triangle vertices are shown by their letters in the grid space they occupy, and their color corresponds to the color of the points on the opposite line which are not vertices. All triangles are shown centered on their centroid.

Two tables for $p = 17$ and $p = 19$ are given so that the reader can confirm that the forms do realize the same spreads. The reader is also invited to confirm that all possible spread triples over each field are reported in each table. Each table was computed by measuring every triangle (up to translation and scaling) mod p .

These examples illustrate that quadratic forms of the same determinantal type realize the same spread triples, even though the distribution across triangles may differ.

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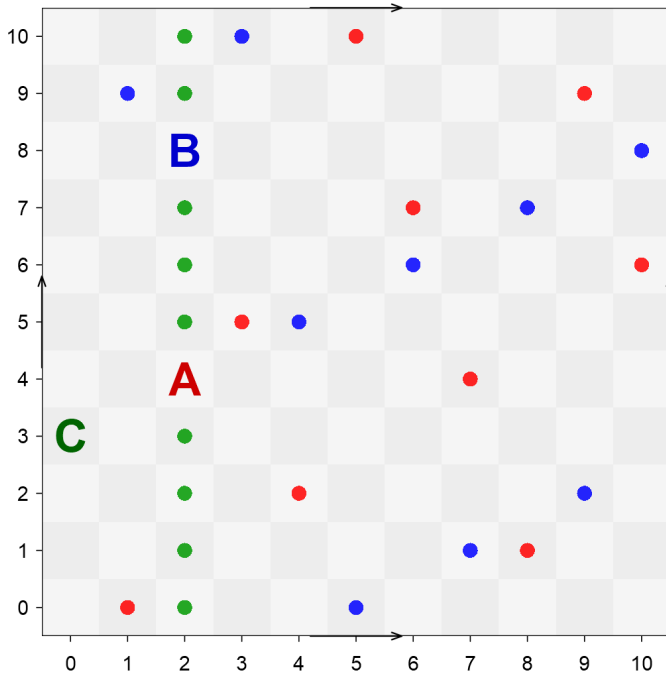
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$\mathcal{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	A	B	C
quadrance opposite	4	5	5
spread at	4	5	5

$\mathcal{R} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	A	B	C
quadrance opposite	3	3	3
spread at	6	6	6

Figure 1: Triangle A=(2,3), B=(0,4), C=(0,2) over $\mathbb{F}_7$



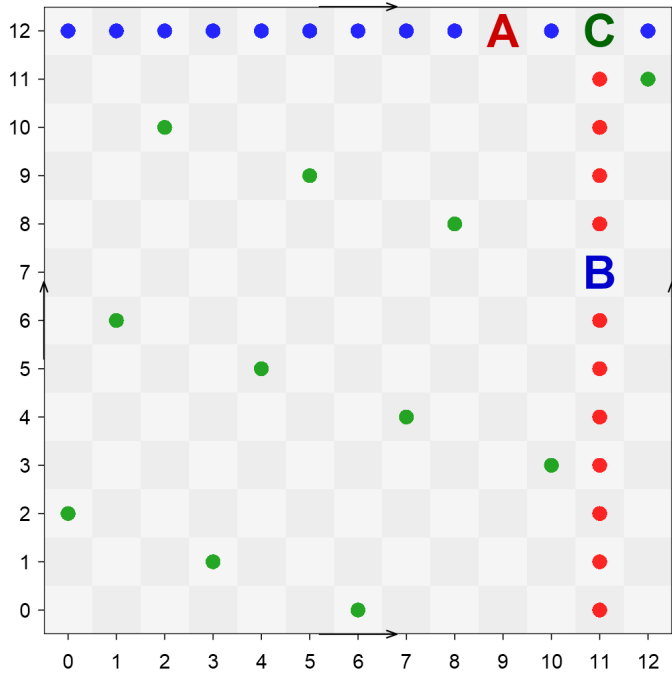
$\mathcal{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	A	B	C
quadrance opposite	7	5	5
spread at	3	10	10

$\mathcal{R} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	A	B	C
quadrance opposite	1	3	6
spread at	5	4	8

Figure 2: Triangle A=(2,4), B=(2,8), C=(0,3) over $\mathbb{F}_{11}$

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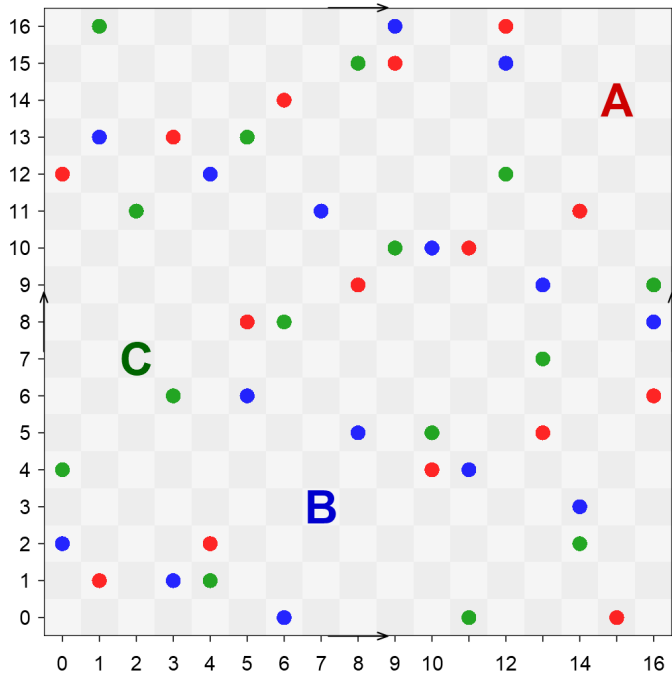
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$\mathcal{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	A	B	C
quadrance opposite	12	4	3
spread at	4	10	1

$\mathcal{R} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	A	B	C
quadrance opposite	1	4	5
spread at	8	6	1

Figure 3: Triangle A=(9,12), B=(11,7), C=(11,12) over $\mathbb{F}_{13}$



$\mathcal{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	A	B	C
quadrance opposite	7	14	15
spread at	12	7	16

$\mathcal{R} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	A	B	C
quadrance opposite	9	1	11
spread at	12	7	9

Figure 4: Triangle A=(15,14), B=(7,3), C=(2,7) over $\mathbb{F}_{17}$

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$A = \begin{bmatrix} 3 & 7 \\ 7 & 7 \end{bmatrix}$				$B = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}$			
(1, 2, 10)	(1, 3, 9)	(1, 6, 6)	(2, 2, 3)	(1, 4, 8)	(1, 5, 7)	(4, 4, 7)	(4, 5, 7)
(2, 3, 6)	(2, 6, 9)	(2, 9, 10)	(3, 3, 9)	(4, 5, 8)	(5, 5, 8)	(7, 7, 8)	(7, 8, 8)
(3, 6, 10)	(3, 10, 10)	(6, 9, 10)	(9, 9, 9)				

Table 1: Over $\mathbb{F}_{11}$, A has $\det(A) \equiv 5 \pmod{11}$, which makes its determinant a quadratic residue modulo 11, and it realizes 12 similarity classes, while B has $\det(B) \equiv 7 \pmod{11}$, which makes its determinant a quadratic non-residue modulo 11, and it realizes 8 similarity classes.

$C = \begin{bmatrix} 10 & 5 \\ 5 & 10 \end{bmatrix}$	$D = \begin{bmatrix} 1 & 9 \\ 9 & 5 \end{bmatrix}$																												
<table style="width: 100%; border-collapse: collapse;"> <tr><td style="border-top: 1px solid black; border-bottom: 1px solid black;">(1, 4, 10)</td><td style="border-top: 1px solid black; border-bottom: 1px solid black;">(1, 6, 8)</td><td style="border-top: 1px solid black; border-bottom: 1px solid black;">(1, 7, 7)</td><td style="border-top: 1px solid black; border-bottom: 1px solid black;">(4, 4, 4)</td></tr> <tr><td style="border-bottom: 1px solid black;">(4, 6, 7)</td><td style="border-bottom: 1px solid black;">(4, 6, 8)</td><td style="border-bottom: 1px solid black;">(4, 7, 8)</td><td style="border-bottom: 1px solid black;">(4, 10, 10)</td></tr> <tr><td style="border-bottom: 1px solid black;">(6, 6, 10)</td><td style="border-bottom: 1px solid black;">(6, 7, 10)</td><td style="border-bottom: 1px solid black;">(7, 8, 10)</td><td style="border-bottom: 1px solid black;">(8, 8, 10)</td></tr> </table>	(1, 4, 10)	(1, 6, 8)	(1, 7, 7)	(4, 4, 4)	(4, 6, 7)	(4, 6, 8)	(4, 7, 8)	(4, 10, 10)	(6, 6, 10)	(6, 7, 10)	(7, 8, 10)	(8, 8, 10)	<table style="width: 100%; border-collapse: collapse;"> <tr><td style="border-top: 1px solid black; border-bottom: 1px solid black;">(1, 2, 12)</td><td style="border-top: 1px solid black; border-bottom: 1px solid black;">(1, 3, 11)</td><td style="border-top: 1px solid black; border-bottom: 1px solid black;">(1, 5, 9)</td><td style="border-top: 1px solid black; border-bottom: 1px solid black;">(2, 2, 5)</td></tr> <tr><td style="border-bottom: 1px solid black;">(2, 3, 3)</td><td style="border-bottom: 1px solid black;">(2, 3, 9)</td><td style="border-bottom: 1px solid black;">(2, 5, 11)</td><td style="border-bottom: 1px solid black;">(2, 9, 12)</td></tr> <tr><td style="border-bottom: 1px solid black;">(2, 11, 11)</td><td style="border-bottom: 1px solid black;">(3, 5, 9)</td><td style="border-bottom: 1px solid black;">(3, 5, 12)</td><td style="border-bottom: 1px solid black;">(3, 11, 12)</td></tr> <tr><td style="border-bottom: 1px solid black;">(5, 5, 11)</td><td style="border-bottom: 1px solid black;">(5, 12, 12)</td><td style="border-bottom: 1px solid black;">(9, 9, 11)</td><td style="border-bottom: 1px solid black;">(9, 11, 12)</td></tr> </table>	(1, 2, 12)	(1, 3, 11)	(1, 5, 9)	(2, 2, 5)	(2, 3, 3)	(2, 3, 9)	(2, 5, 11)	(2, 9, 12)	(2, 11, 11)	(3, 5, 9)	(3, 5, 12)	(3, 11, 12)	(5, 5, 11)	(5, 12, 12)	(9, 9, 11)	(9, 11, 12)
(1, 4, 10)	(1, 6, 8)	(1, 7, 7)	(4, 4, 4)																										
(4, 6, 7)	(4, 6, 8)	(4, 7, 8)	(4, 10, 10)																										
(6, 6, 10)	(6, 7, 10)	(7, 8, 10)	(8, 8, 10)																										
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(2, 3, 3)	(2, 3, 9)	(2, 5, 11)	(2, 9, 12)																										
(2, 11, 11)	(3, 5, 9)	(3, 5, 12)	(3, 11, 12)																										
(5, 5, 11)	(5, 12, 12)	(9, 9, 11)	(9, 11, 12)																										

Table 2: Over $\mathbb{F}_{13}$, C has $\det(C) \equiv 10 \pmod{13}$, which makes its determinant a quadratic residue modulo 13, and it realizes 12 similarity classes, while D has $\det(D) \equiv 2 \pmod{13}$, which makes its determinant a quadratic non-residue modulo 13, and it realizes 16 similarity classes.

$E = \begin{bmatrix} 11 & 9 \\ 9 & 5 \end{bmatrix}$				$F = \begin{bmatrix} 11 & 11 \\ 11 & 10 \end{bmatrix}$			
(1, 2, 16)	(1, 6, 12)	(1, 7, 11)	(1, 9, 9)	(1, 3, 15)	(1, 4, 14)	(1, 5, 13)	(1, 8, 10)
(2, 2, 9)	(2, 6, 7)	(2, 6, 12)	(2, 7, 7)	(3, 3, 10)	(3, 4, 4)	(3, 4, 13)	(3, 5, 10)
(2, 9, 16)	(2, 11, 11)	(2, 11, 12)	(6, 6, 16)	(3, 5, 14)	(3, 8, 13)	(3, 8, 15)	(3, 14, 14)
(6, 7, 9)	(6, 9, 11)	(6, 11, 16)	(7, 9, 12)	(4, 5, 8)	(4, 5, 15)	(4, 8, 10)	(4, 10, 13)
(7, 11, 16)	(7, 12, 16)	(9, 11, 12)	(9, 16, 16)	(4, 14, 15)	(5, 5, 5)	(5, 8, 15)	(5, 10, 14)
(12, 12, 16)				(5, 13, 13)	(8, 8, 14)	(8, 13, 14)	(10, 10, 14)
				(10, 13, 15)	(10, 15, 15)	(13, 14, 15)	

Table 3: Over $\mathbb{F}_{17}$, E has $\det(E) \equiv 8 \pmod{17}$, which makes its determinant a quadratic residue modulo 17, and it realizes 21 similarity classes, while F has $\det(F) \equiv 6 \pmod{17}$, which makes its determinant a quadratic non-residue modulo 17, and it realizes 27 similarity classes.

Counting similar triangles with coordinates in finite fields

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$G = \begin{bmatrix} 14 & 3 \\ 3 & 4 \end{bmatrix}$				$H = \begin{bmatrix} 8 & 15 \\ 15 & 15 \end{bmatrix}$			
(1, 2, 16)	(1, 6, 12)	(1, 7, 11)	(1, 9, 9)	(1, 3, 15)	(1, 4, 14)	(1, 5, 13)	(1, 8, 10)
(2, 2, 9)	(2, 6, 7)	(2, 6, 12)	(2, 7, 7)	(3, 3, 10)	(3, 4, 4)	(3, 4, 13)	(3, 5, 10)
(2, 9, 16)	(2, 11, 11)	(2, 11, 12)	(6, 6, 16)	(3, 5, 14)	(3, 8, 13)	(3, 8, 15)	(3, 14, 14)
(6, 7, 9)	(6, 9, 11)	(6, 11, 16)	(7, 9, 12)	(4, 5, 8)	(4, 5, 15)	(4, 8, 10)	(4, 10, 13)
(7, 11, 16)	(7, 12, 16)	(9, 11, 12)	(9, 16, 16)	(4, 14, 15)	(5, 5, 5)	(5, 8, 15)	(5, 10, 14)
(12, 12, 16)				(5, 13, 13)	(8, 8, 14)	(8, 13, 14)	(10, 10, 14)
				(10, 13, 15)	(10, 15, 15)	(13, 14, 15)	

Table 4: Over $\mathbb{F}_{17}$, G has $\det(G) \equiv 13 \pmod{17}$, which makes its determinant a quadratic residue modulo 17, and it realizes 21 similarity classes, while H has $\det(H) \equiv 14 \pmod{17}$, which makes its determinant a quadratic non-residue modulo 17, and it realizes 27 similarity classes.

$M = \begin{bmatrix} 15 & 14 \\ 14 & 18 \end{bmatrix}$				$N = \begin{bmatrix} 6 & 17 \\ 17 & 15 \end{bmatrix}$			
(1, 2, 18)	(1, 4, 16)	(1, 8, 12)	(1, 9, 11)	(1, 3, 17)	(1, 5, 15)	(1, 6, 14)	(1, 7, 13)
(1, 10, 10)	(2, 2, 11)	(2, 4, 8)	(2, 4, 10)	(3, 3, 14)	(3, 5, 6)	(3, 5, 7)	(3, 6, 17)
(2, 8, 9)	(2, 9, 18)	(2, 10, 16)	(2, 11, 12)	(3, 7, 7)	(3, 13, 13)	(3, 13, 15)	(3, 14, 15)
(2, 12, 16)	(4, 4, 9)	(4, 8, 8)	(4, 9, 11)	(5, 5, 15)	(5, 6, 13)	(5, 7, 14)	(5, 13, 17)
(4, 10, 18)	(4, 11, 16)	(4, 12, 12)	(4, 12, 18)	(5, 14, 17)	(6, 6, 13)	(6, 7, 14)	(6, 7, 15)
(8, 9, 10)	(8, 10, 11)	(8, 11, 18)	(8, 12, 16)	(6, 15, 17)	(7, 13, 17)	(7, 15, 17)	(13, 14, 14)
(8, 16, 18)	(9, 9, 16)	(9, 10, 12)	(9, 12, 18)	(13, 14, 15)	(14, 17, 17)	(15, 15, 15)	
(9, 16, 16)	(10, 11, 12)	(10, 16, 18)	(11, 11, 16)				
(11, 18, 18)							

Table 5: Over $\mathbb{F}_{19}$, M has $\det(M) \equiv 17 \pmod{19}$, which makes its determinant a quadratic residue modulo 19, and it realizes 33 similarity classes, while N has $\det(N) \equiv 10 \pmod{19}$, which makes its determinant a quadratic non-residue modulo 19, and it realizes 27 similarity classes.

$P = \begin{bmatrix} 1 & 9 \\ 9 & 9 \end{bmatrix}$				$Q = \begin{bmatrix} 0 & 14 \\ 14 & 15 \end{bmatrix}$			
(1, 2, 18)	(1, 4, 16)	(1, 8, 12)	(1, 9, 11)	(1, 3, 17)	(1, 5, 15)	(1, 6, 14)	(1, 7, 13)
(1, 10, 10)	(2, 2, 11)	(2, 4, 8)	(2, 4, 10)	(3, 3, 14)	(3, 5, 6)	(3, 5, 7)	(3, 6, 17)
(2, 8, 9)	(2, 9, 18)	(2, 10, 16)	(2, 11, 12)	(3, 7, 7)	(3, 13, 13)	(3, 13, 15)	(3, 14, 15)
(2, 12, 16)	(4, 4, 9)	(4, 8, 8)	(4, 9, 11)	(5, 5, 15)	(5, 6, 13)	(5, 7, 14)	(5, 13, 17)
(4, 10, 18)	(4, 11, 16)	(4, 12, 12)	(4, 12, 18)	(5, 14, 17)	(6, 6, 13)	(6, 7, 14)	(6, 7, 15)
(8, 9, 10)	(8, 10, 11)	(8, 11, 18)	(8, 12, 16)	(6, 15, 17)	(7, 13, 17)	(7, 15, 17)	(13, 14, 14)
(8, 16, 18)	(9, 9, 16)	(9, 10, 12)	(9, 12, 18)	(13, 14, 15)	(14, 17, 17)	(15, 15, 15)	
(9, 16, 16)	(10, 11, 12)	(10, 16, 18)	(11, 11, 16)				
(11, 18, 18)							

Table 6: Over $\mathbb{F}_{19}$, P has $\det(Q) \equiv 4 \pmod{19}$, which makes its determinant a quadratic residue modulo 19, and it realizes 33 similarity classes, while Q has $\det(Q) \equiv 13 \pmod{19}$, which makes its determinant a quadratic non-residue modulo 19, and it realizes 27 similarity classes.